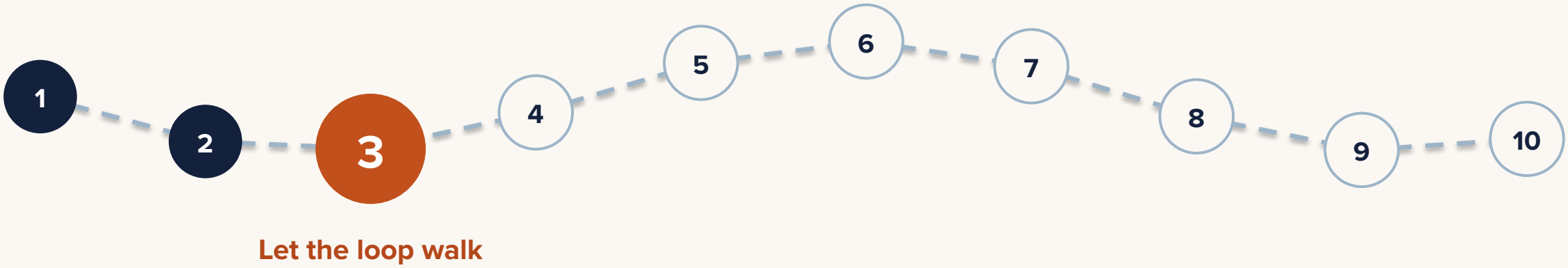
PYTHON FOR SUPPLY CHAIN

Session 3 · Mon 23 Nov 2026

Let the loop walk

for loops, functions, return, tracebacks



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Today

1

for loops

`for q in quantities:`

2

Functions

`def order_size(q):`

3

return

`hand it back`

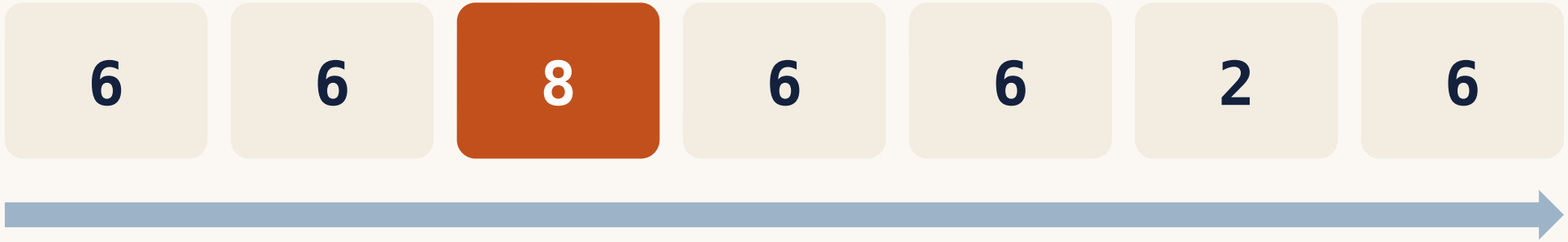
4

Tracebacks

`read from the bottom`

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A loop walks down the shelf




```
for quantity in quantities:
    print(quantity)
```

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The indent is the loop body

```
for quantity in quantities:
    double = quantity * 2
    print(double)
print("done")
```

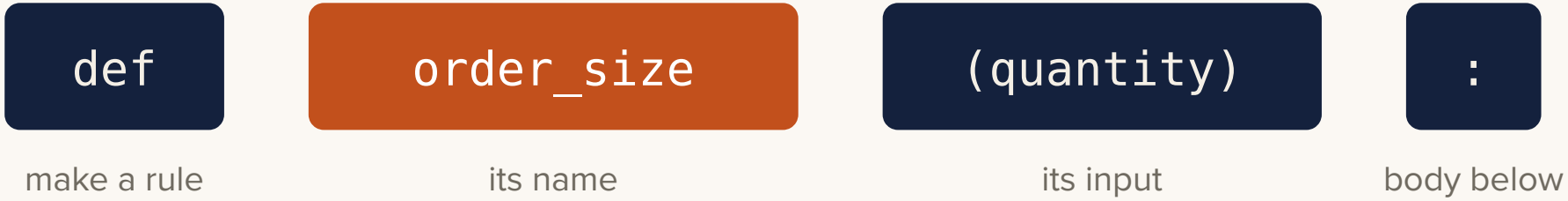
7 times

once

Indented: every turn. Not indented: after the loop.

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A function is your own rule

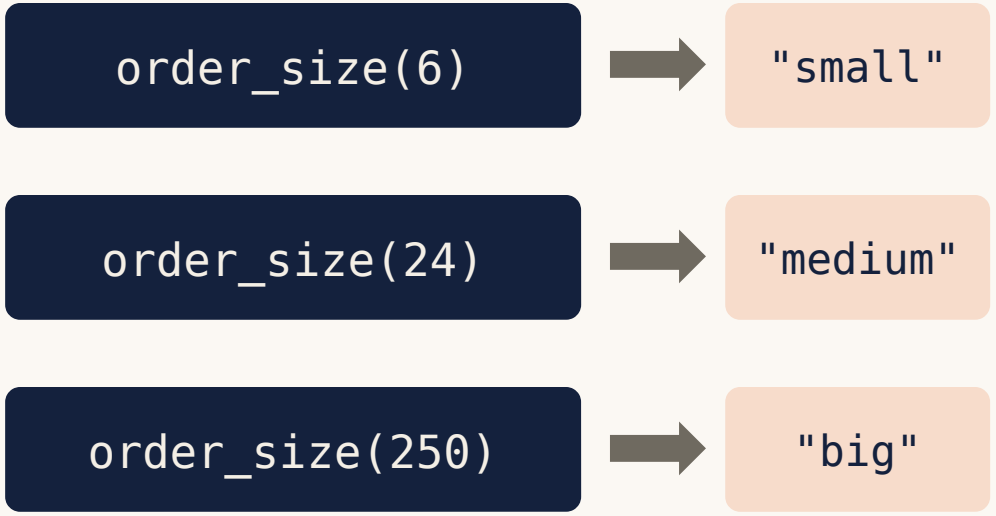


Writing def runs nothing. Call it to use it.

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Write once, call many times

```
def order_size(quantity):
    if quantity >= 100:
        return "big"
    elif quantity >= 12:
        return "medium"
    else:
        return "small"
```



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return hands the answer back

return

```
def size_a(quantity):
    return "small"

answer = size_a(6)
print(answer)
```

```
→ small
```

print only

```
def size_b(quantity):
    print("small")

answer = size_b(6)
print(answer)
```

```
→ small
→ None
```

No return: the answer is None.

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A function can take two inputs



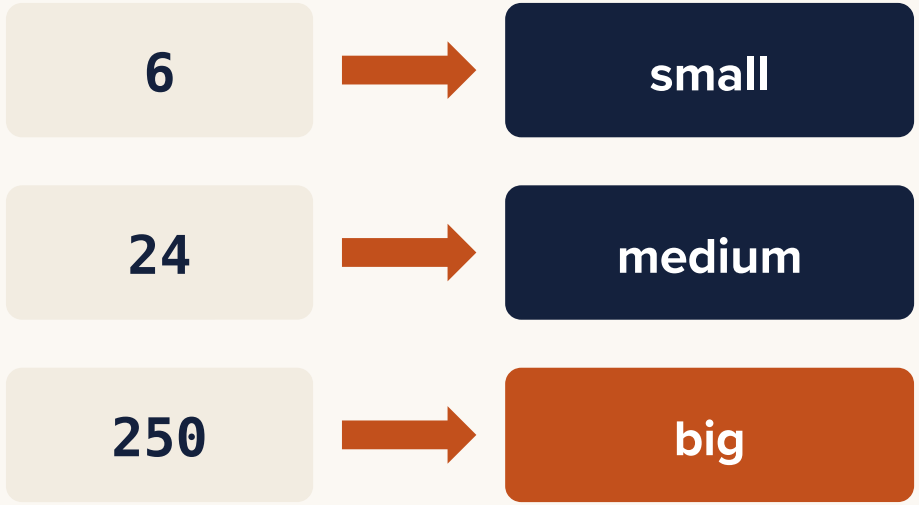

```
line_value(6, 2.55)
→ 15.3
```

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Loop + function: label every line

```
for quantity in [6, 24, 250]:
    label = order_size(quantity)
    print(f"{quantity} units -> {label}")
```

```
→ 6 units -> small
→ 24 units -> medium
→ 250 units -> big
```



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A list of product cards

WHITE HANGING HEART...

quantity 6

price 2.55

WHITE METAL LANTERN

quantity 6

price 3.39

CREAM CUPID HEARTS...

quantity 8

price 2.75

```
for line in order_lines:
    print(line["quantity"])
```

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A traceback is a stack of plates

```
for quantity in mixed_quantities:
```

the loop calls

```
print(order_size(quantity))
```

the function

```
if quantity >= 100:
```

where it broke

```
TypeError: '>=' not supported between 'str' and 'int'
```

read the bottom line first, then go up

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Three mistakes you will meet

`IndentationError`

the body is not indented

```
for q in quantities:
        print(q)
```

`None`

the function has no return

```
answer = size_b(6)
```

`NameError`

called before def ran

```
line_value(6, 2.55)
```

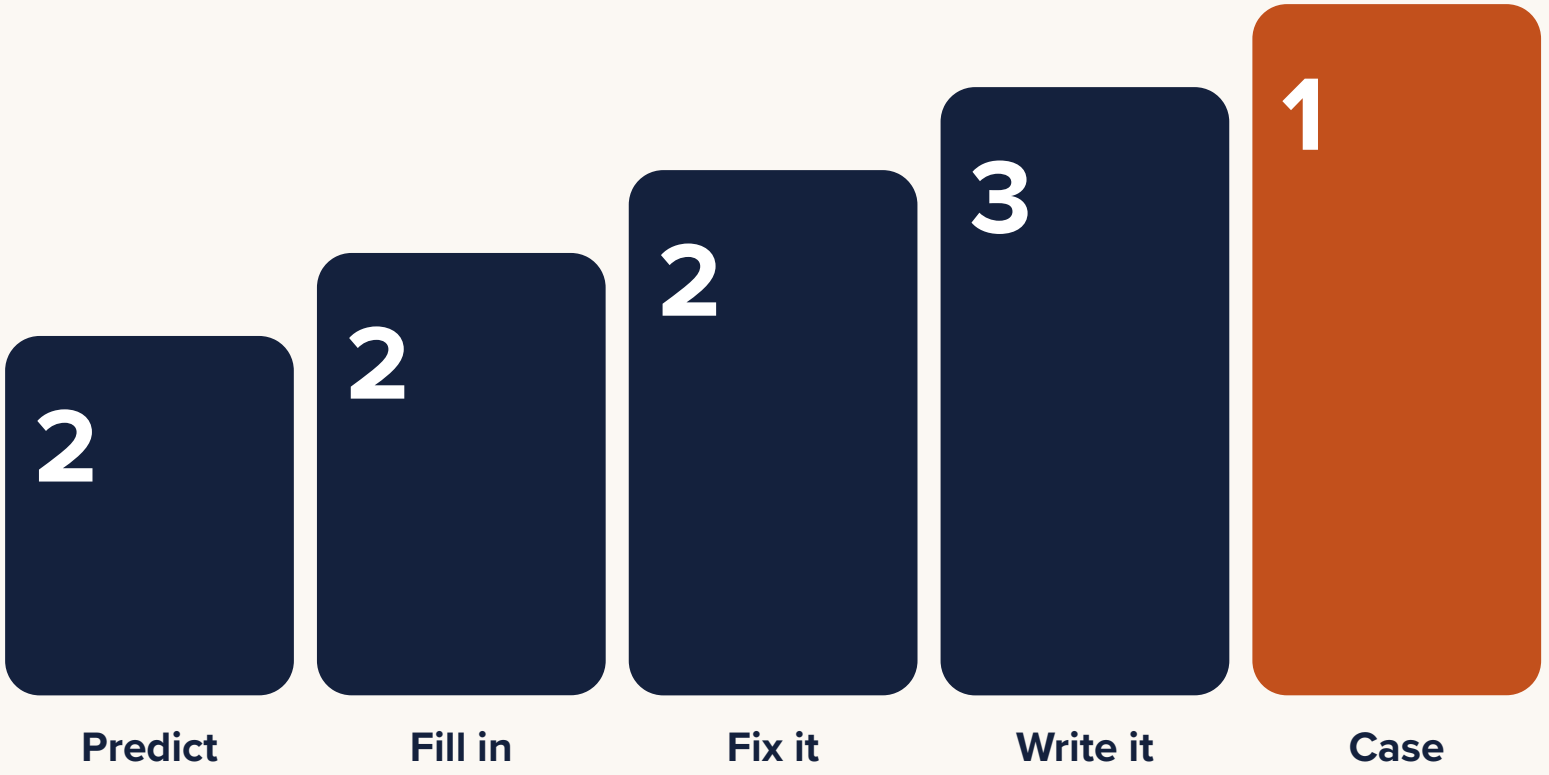
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Your toolkit so far

Code	Does
`for x in a_list:`	repeat for every value
`    (4 spaces)`	the loop or rule body
`def name(input):`	make your own rule
`return value`	hand the answer back
`name(6)`	call the rule
`None`	no answer came back
`traceback`	read from the bottom

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Now in Colab



Notebook

s3_let_loop_walk

Homework

2 extras, mount Drive

Recall

Tue 24, Wed 25

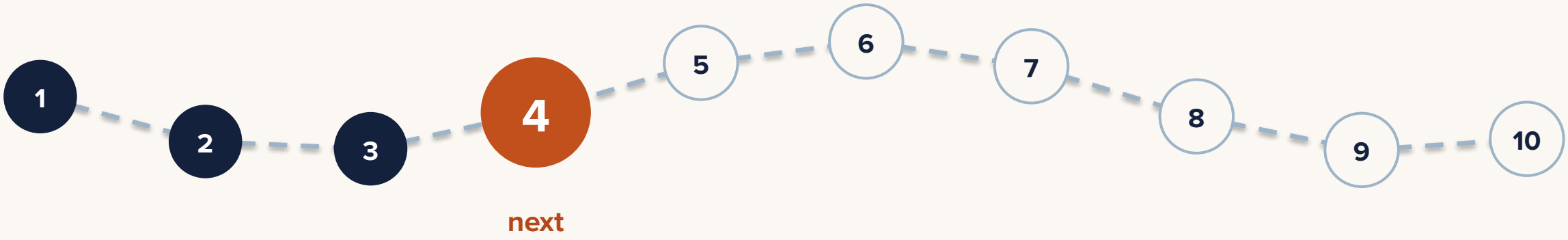
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Thank you

Next: Session 4

What does clean data look like?

Thu 26 Nov



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